



UIST

User Interface Software & Technology Presentation

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INF 232



Overview

Day

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Core Topics



Novel Interaction Techniques

- New input/output methods (gesture, touch, mid-air, pen, haptics).

Input & Output Devices

- Sensors, actuators, wearables, displays, tangible hardware.

Interactive Systems & Prototypes

- Technically novel, fully implemented interaction systems.

Toolkits & Interaction Frameworks

- Software/hardware platforms enabling new interactions.

Sensing & ML for Interaction

- Vision, audio, physiological sensing, signal processing for I/O.

AI-Enhanced Interaction (when interaction-driven)

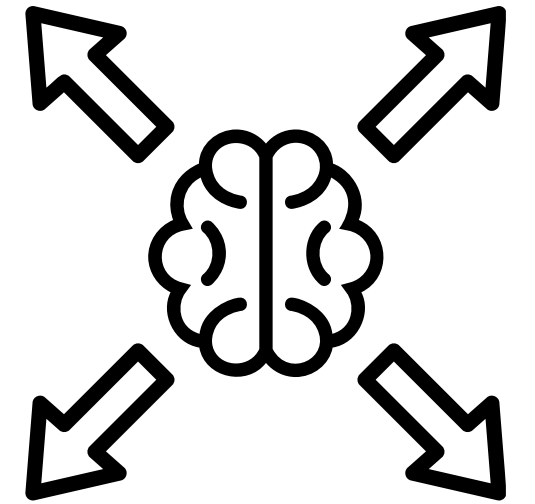
- AI as an enabler for new interaction techniques (not AI models alone).

Core Research Questions: The "How" and "Why"



- How can we integrate multiple technical components, hardware, algorithms, human computation, and interaction techniques, to enable and demonstrate novel interactive capabilities?
- How do we design tools, libraries, and languages that support the construction, engineering, or validation of interactive systems by the broader research community?
- How can we engineer interaction techniques that allow users to ignore the boundaries between different computers in networking environment?

Disciplines Outside of HCI



- Engineering (Electrical, Mechanical, Biomedical)
- Computer Science (Systems, Graphics, Vision, Robotics)
- Materials Science
- Data Science & Machine Learning
- Design & Architecture
- Cognitive Science & Psychology

Venues

- **CHI** – *The biggest and most prestigious HCI conference; highly competitive; covers all HCI topics*
- **UIST** – *Top venue specifically for novel user interfaces, interaction techniques, and enabling systems*
- **TEI** – *Popular for tangible and embodied interaction; well-known in physical computing and wearable communities*
- **ICMI** – *Highly respected for multimodal interaction research (speech, gesture, gaze)*
- **VR/AR / IEEE VR & ISMAR** – *Popular for immersive and 3D interfaces; strong community in VR/AR research*

Developing Novel Devices: Hardware, Materials, and Fabrication

Typical contributions:

- New sensing and tracking devices
- Display technology
- Haptics feedback devices
- Robotic approaches
- Developments in materials that lead to novel interactive capabilities
- New fabrication techniques

Interacting with Devices: Interaction Techniques & Modalities

Typical contributions:

- Novel interaction techniques (touch, gesture, speech, gaze)
- Improved usability, performance, or accessibility of existing modalities
- Cross-device and cross-context interactions (handheld, wearable, 2D/3D, tangibles)
- Clear design rationale grounded in user needs
- Evaluations showing improvements over existing techniques
- New opportunities for interaction through novel modalities or combinations

Blending Interaction: Engineering Interactive Systems & Tools

Typical contributions:

- Development of novel interactive systems combining hardware, algorithms, and interaction techniques
- “Enabling” contributions: datasets, tools, libraries, infrastructure, or languages
- Systems and resources should demonstrate new interactive capabilities
- Contributions should support construction, engineering, or validation of interactive systems
- Emphasis on shareability to benefit the research community
- Inspire future interface design and interactive system explorations

Venue Comparisons: Scope & Calibration

Core Similarities: High-Level Research Standards

While the specific focus areas change, UIST and the technical CHI subcommittees share foundational archival standards:

- All focus on human-computer interaction and advancing interface design.*
- Emphasis on novelty and contribution, new techniques, systems, or methods that improve or enable interactions.*
- Require strong design rationale and often evaluation to show improvements or capabilities.*
- Encourage work that inspires future research or interface designs.*

Scoping Differences: UIST vs. CHI Subcommittees

Subcommittee	Focus	Typical Contributions	Scope
UIST	Broader venue for novel systems, software, and tools for user interfaces	Interactive systems, frameworks, toolkits, or platforms; both hardware and software integration	Hardware + software, high-level enabling contributions; “big picture” UI technologies (foundational infrastructure and architectural models)
CHI: Developing Novel Devices	Creation of new hardware or devices	Novel input/output devices, sensors, tangible objects	Physical device innovation; prototypes of devices enabling new ways to interact
CHI: Interacting with Devices	How people interact with devices	New interaction techniques or modalities (touch, gesture, gaze, haptics, physiology)	Focus on interaction design, not necessarily creating new devices; can be software or hardware-based
CHI: Blending Interaction	Combining multiple interaction modalities	Techniques that mix modalities (e.g., speech + gesture, touch + haptics)	Focus on multi-modal interaction; integration of modalities to enhance experience

UIST Timeline

- **1988 – Inception**
 - *First UIST held*
 - *Focus on graphical user interfaces, windowing systems, and software toolkits*
- **1990s – Interaction Techniques Emerge**
 - *Research on direct manipulation, pen-based computing, and early haptics*
 - *Expansion into user interface toolkits and early prototyping systems*
- **2000s – Ubiquitous & Tangible Interfaces**
 - *Inclusion of mobile, ubiquitous computing, and tangible interaction*
 - *Focus on novel interaction devices and multi-sensor systems*
- **2010s – Multi-Modal & System Integration**
 - *Growth in hardware-software integrated systems*
 - *Emphasis on multi-modal interactions (touch, gesture, gaze, haptics)*
 - *Rise of enabling technologies (libraries, toolkits, datasets)*
- **2020s – Cutting-Edge & Enabling Research**
 - *Leading venue for novel interactive systems and enabling contributions*
 - *Strong focus on evaluated, practical, and shareable systems*
 - *Encourages research that inspires future interface designs*

UIST & CHI: Historical Timeline

1982–1987: Shared Roots

- All technical UI work was submitted to CHI.
- Engineers felt “building” work (toolkits, prototypes) was undervalued compared to user studies.
- Led to the creation of UIST in 1988 as a venue for technical and systems-focused research.

1990s: Parallel Growth

- ACM SIGCHI became the parent organization for both CHI and UIST.
- Researchers often demoed at UIST (Fall) and published analytical/user-study work at CHI (Spring).

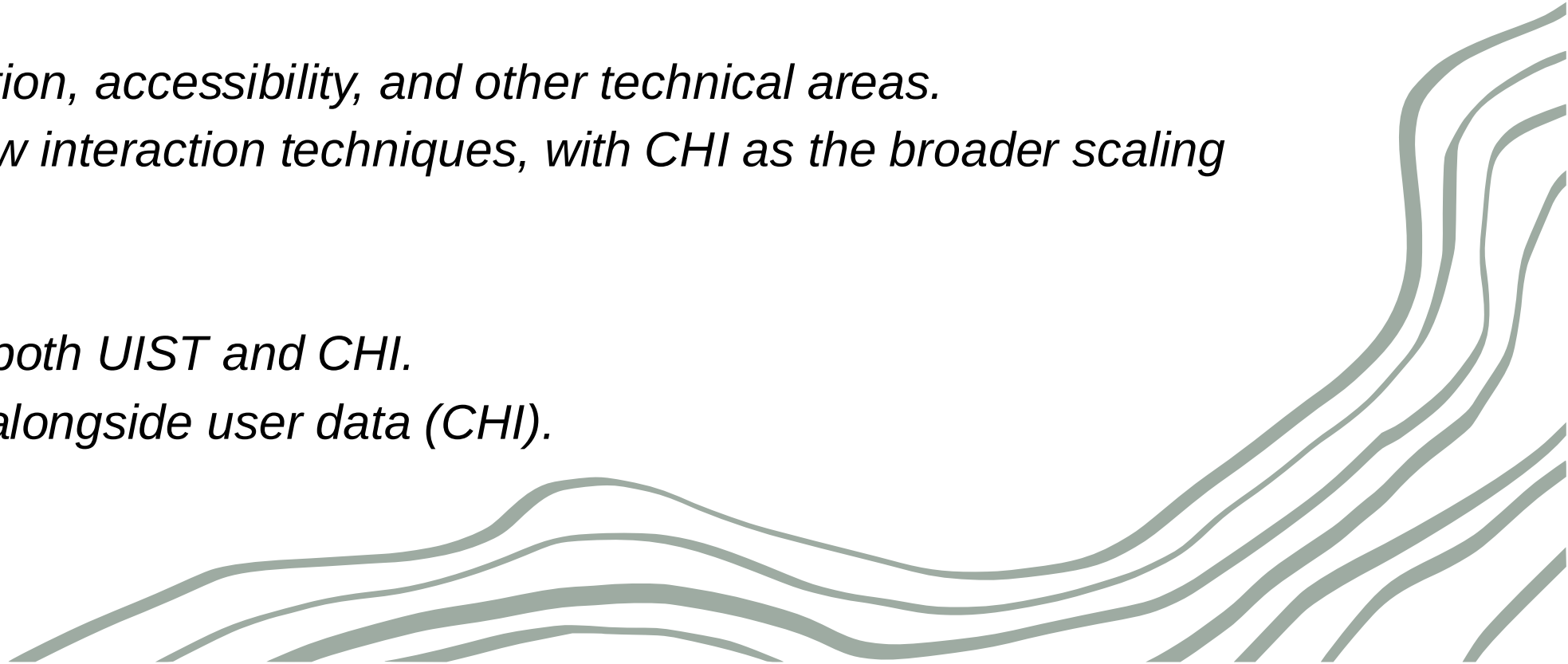
2000s: CHI Subcommittees

- CHI grew very large; technical papers were sometimes rejected by reviewers focused on user studies.
- Subcommittees were introduced to ensure proper review of engineering-heavy work.
- UIST veterans helped staff technical subcommittees, bridging UIST-style research to CHI.

2010s: Cross-Pollination

- *Shared focus groups emerged on haptics, fabrication, accessibility, and other technical areas.*
- *UIST became recognized as an “incubator” for new interaction techniques, with CHI as the broader scaling venue.*

2020s: Unified Identity & Open Science

- *Program chairs and reviewers increasingly serve both UIST and CHI.*
 - *Emphasis on open science: sharing code (UIST) alongside user data (CHI).*
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Evolution of Contribution

- **1988:** Focused on novel systems and prototypes; evaluation was mostly informal demos and concept validation.
 - **1990s:** Added structured usability tests and small-scale user studies; contributions judged on technical innovation and potential impact.
 - **2000s:** Emphasis on feasibility and utility of complex systems; evaluation methods included task performance, observations, and pilot deployments.
 - **2010s:** Strong expectation for formal evaluation and clear design rationale; enabling contributions (toolkits, libraries, datasets) judged on reusability and influence on future research.
 - **2020s:** Focus on rigor, reproducibility, open science, and community impact; contributions judged on novelty, empirical validation, and potential to advance the field.
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Past, Present, and Future of User Interface Software Tools

Overview: *A retrospective analysis of how research-born tools like window managers and toolkits became industry standards, while forecasting a shift toward ubiquitous computing and recognition-based interfaces*

Takeaway: *Future tools must move beyond discrete "event-based" models (clicks) to architectures organized around rich user context and uncertain inputs like speech and gesture*

Tangible Bits: Towards Seamless Interfaces between People, Bits and Atoms

Overview: *A foundational vision to bridge the gap between cyberspace (bits) and the physical environment (atoms) by leveraging human senses and physical skills neglected by traditional screen-based GUIs.*

Takeaway: *By making digital information tangible and peripheral, we can turn the "world itself into an interface," reducing the cognitive load of constant foreground focus*

Pick-and-Drop: Direct Manipulation for Multiple Computers

Overview: Introduces an interaction technique allowing users to transfer data between different computers (e.g., from a PDA to a wall display) as if they were physically moving an object.

Takeaway: Adding "physicalness" to virtual worlds makes multi-device interaction visible and understandable, which is crucial for seamless collaboration in ubiquitous environments.



Discussion Day

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Discussion of UIST

**VisCourt: In-Situ Guidance for Interactive Tactic Training in
Mixed Reality (UIST '24)**
**ProForm: Solder-Free Circuit Assembly Using Thermoforming
(UIST '25).**

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VisCourt (UIST '24)

Why We Chose This Paper:

- It's about basketball! :)
- It exemplifies a complex integrated system that combines hardware (MR headsets), computer vision (Mocap), and specialized interaction logic to enable a novel human capability: individual team-level practice

<https://youtu.be/Ez9iaXrU3Jk?si=ekTFpFJII-Qtjm3V>

VisCourt: Quick Technical Overview

- **The Goal:** To bridge the "comprehension gap" where athletes understand 2D tactical theory but struggle with 3D on-court coordination.
- **System Implementation:**
 - **Pipeline:** Uses seven GoPro cameras and EasyMocap for multi-person 3D reconstruction.
 - **Interaction:** Employs "Bullet Time" (play/pause/rewind via user movement) and ghost avatars to show tactical "What-Ifs".
 - **Hardware:** Unity 2021.3.15 and PICO4 MR HMD.
- **Methods:** The authors used multi-view motion capture (Mocap) to create 3D tactical scenarios, developed an iterative design space for in-situ guidance, and conducted user studies with 12 players across varying skill levels.

VisCourt: Argument for Worthiness of Study

- **The Problem:** Existing SportsXR systems are reductionist, focusing on individual motor skills (like free-throws) while ignoring team-level situational awareness.
- **Tactical Complexity:** Basketball tactics are linear in execution but dynamic in response; players often fail because they transform 2D imagery into 3D coordination incorrectly.
- **Safety & Perception:** Traditional VR is unsafe for large-scale movements, while AR has a limited Field of View (FOV). This necessitates a specialized MR engineering solution.

VisCourt: Inquiry and Expansion

Internal HCI Inquiry: Draws from Immersive Analytics and Situated Visualization.

- **External Inquiry:** Incorporates Sports Science (tactical coaching) and Computer Vision (skeleton-based action recognition).
- **Technical Expansion:** The paper advances "Situated Visualization in Motion". Historically, visualizations were stationary; VisCourt engineers guidance for "relative motion" where both the viewer and the virtual players move simultaneously.

VisCourt: Contribution Type & Takeaways

- **Contribution Type:** Artifact, System/Tool.
- **Why Appropriate:** It demonstrates novel interactive capabilities, allowing a single player to simulate team-level coordination without a coach or teammates present.
- **HCI Takeaway:** "Bullet Time" is a highly intuitive metaphor for sports training, allowing users to control time progress through physical position.
- **Interdisciplinary Takeaway:** MR can "quantify experience" by measuring response times to situational "What-If" scenarios in a repeatable environment.

ProForm: Quick Technical Overview

- **The Goal:** A solder-free prototyping method that enables component reuse to combat the e-waste crisis.
- **Method Implementation:**
 - Process: Assembling SMD components on a substrate using Z-tape, then pressure-forming a PETG thermoplastic layer (160°C at 58+ psi).
 - Reversibility: The PETG layer is trimmed away to recover functional components for new projects.
 - Performance: ProFormed circuits maintained electrical integrity for 152+ days and survived 200cm drop tests.

<https://www.youtube.com/watch?v=4lYUpHs-PVs>

ProForm: Argument for Worthiness of Study

- **Sustainability Crisis:** 62 billion kg of e-waste was generated in 2022; current electronics rely on semi-permanent soldering which destroys components during disassembly.
- **The Prototyping Barrier:** Traditional soldering is impractical for unconventional substrates like paper, curved surfaces, or flexible textiles.
- **Enabling Reuse:** There is a critical need for a "reversible manufacturing" tool that is scalable and does not require unique 3D-printed housings for every layout.

ProForm: Inquiry and Expansion

- **Internal HCI Inquiry:** Builds on Sustainable Prototyping and DIY Fabrication (e.g., modular electronics).
- **External Inquiry:** Draws from Manufacturing Engineering (thermoforming) and Materials Science (PETG properties).
- **Technical Expansion:** Expands on "Reversible Manufacturing" by repurposing an industrial technology to encapsulate rather than bond electronics, enabling circuits on flexible and non-planar geometries.



ProForm: Contribution Type & Takeaways

- **Contribution Type:** Artifact/Method (Enabling Contribution).
- **Why Appropriate:** It provides an "enabling resource" (a method and toolkit) that helps other researchers construct and validate future systems, especially those requiring waterproof or flexible properties.
- **HCI Takeaway:** PETG provides optical transparency for debugging while maintaining the mechanical stability needed for non-planar substrates.
- **Interdisciplinary Takeaway:** The environmental impact of producing chips far exceeds the cost of the PETG used for protection; therefore, chip reuse is a major net environmental gain.



More Vidoes

<https://www.youtube.com/watch?v=SeGSaLR5GuE&list=PLqhXYFYmZ-Vd5HOR3d0aW7KNK5Hx2NDC9&index=5>

<https://youtu.be/QCUIg61KAb0?si=aXpn2EfZiiUD0hqu>

https://youtu.be/8F-MEuwspIM?si=4oy8gG_eAYnegaxA

https://youtu.be/2tiAaBN8Uw8?si=9QSzoNw2L__mkWrr

<https://www.youtube.com/watch?v=6nnv4nZhqVg&list=PLqhXYFYmZ-Vd5HOR3d0aW7KNK5Hx2NDC9&index=8>

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Thank You

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